

HABUILD | DATA ANALYST CASE STUDY

Olist Brazilian E-Commerce Public Dataset – Data Story

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EXECUTIVE SUMMARY

The Olist dataset (Sep 2016 – Oct 2018) contains 100k orders across a multi-seller Brazilian marketplace. This analysis uncovers key customer behaviors, delivery performance issues, and revenue drivers to guide customer acquisition, retention, and operational improvements.

HEADLINE METRICS

- Total Orders: 99,441
- Total Revenue: R\$ 16.0 million
- Unique Customers: 96,000
- Repeat Purchase Rate: 3.0% (very low)
- Average Order Value (AOV): R\$160
- Average Review Score: 4.08/5

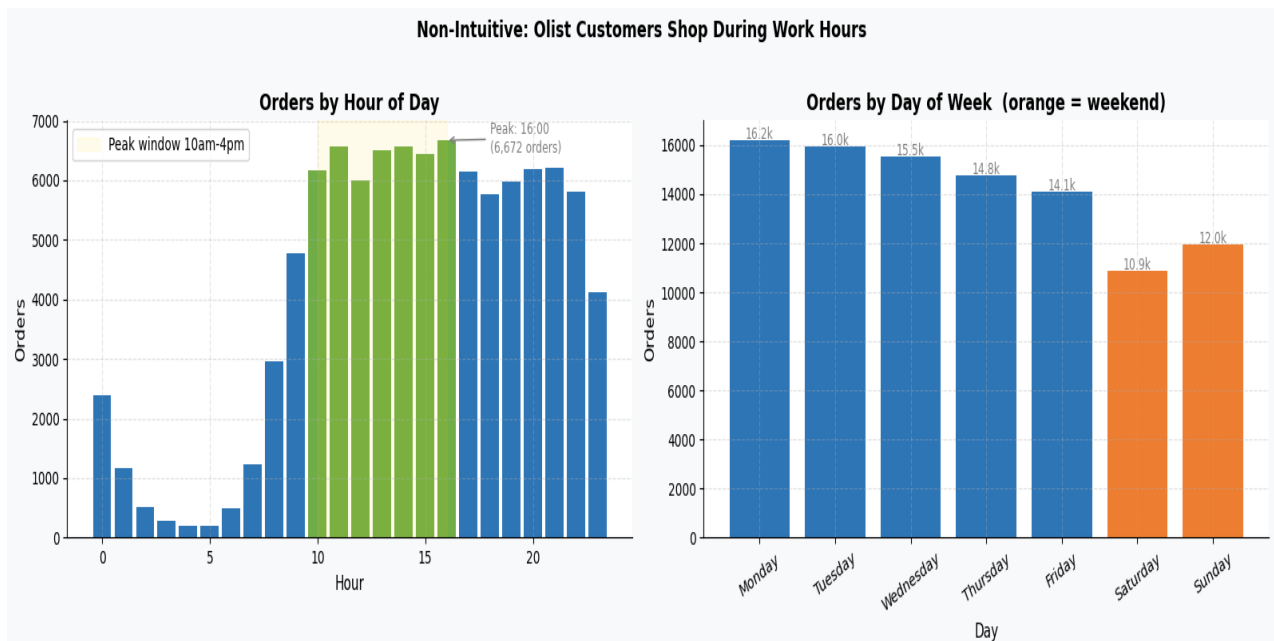
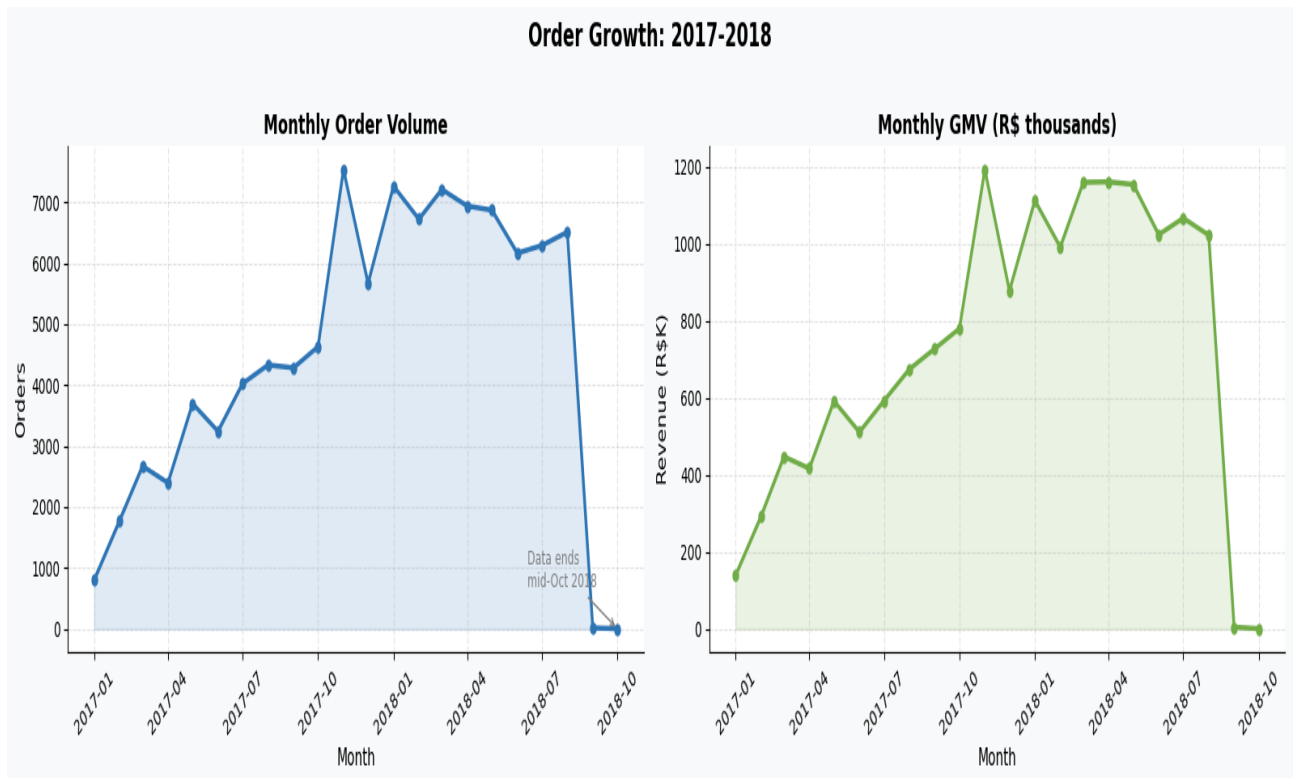
The analysis reveals strong seasonality, significant delivery challenges, and clear opportunities to improve customer lifetime value through better logistics and targeted marketing.

EXPLORATORY DATA ANALYSIS (EDA)

1. Growth Trends & Temporal Patterns

- Orders grew strongly through 2017 and stabilized in 2018. A non-intuitive pattern emerged: while order volume peaks on weekdays, the highest Average Order Value occurs on weekends. Customers buy more deliberately (higher spend) when they have more time.

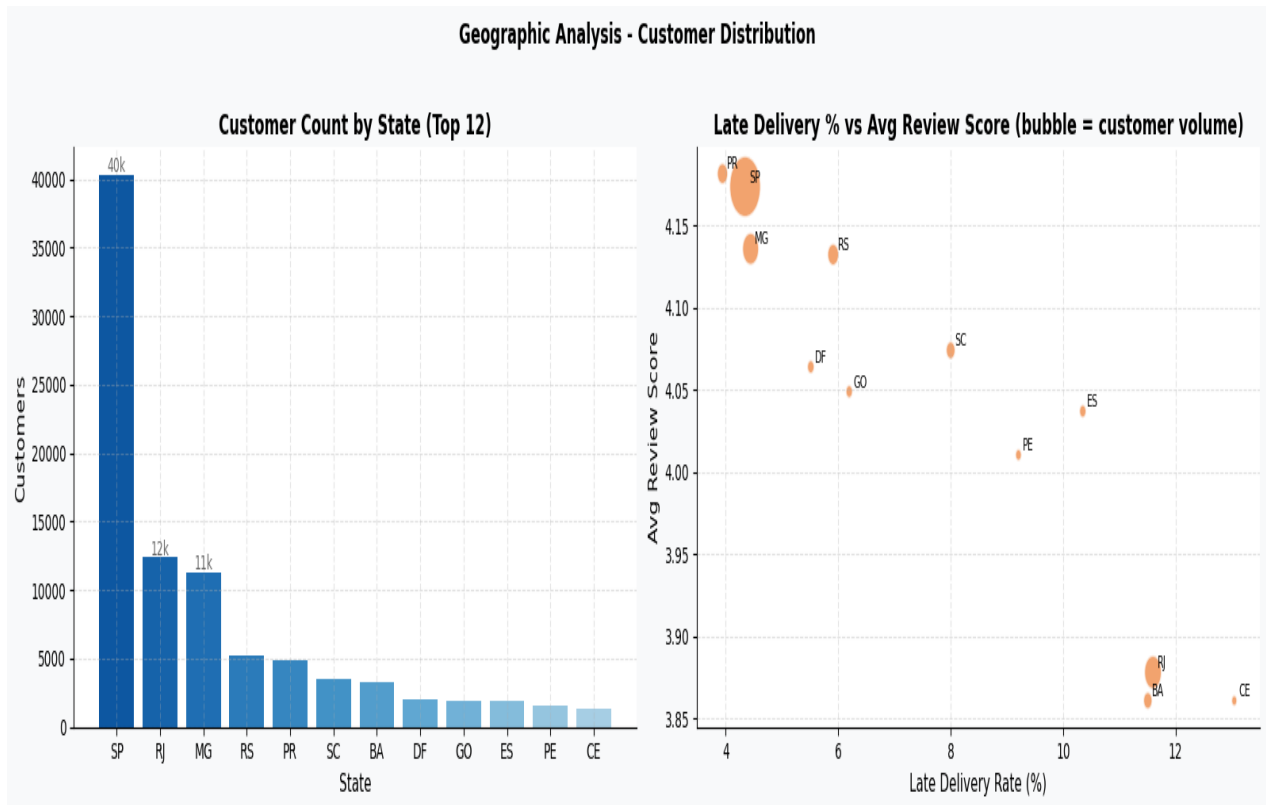
- **Peak ordering hours** are consistently between **10:00–16:00** across all days indicating Olist is used as a purposeful shopping platform rather than impulse/late-night browsing.



2. Geographic Insights

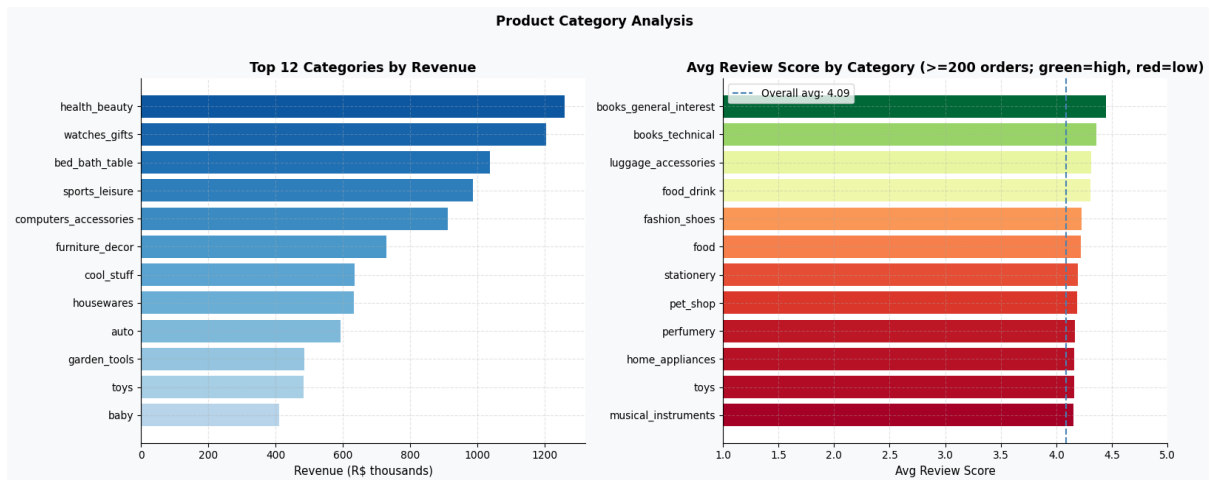
- **São Paulo (SP)** dominates both orders and revenue.

- States with highest average review scores (AP, TO) have very low volume, while high-volume states like SP, RJ, and MG show average scores.
- Delivery times vary dramatically by state some states average >20 days.



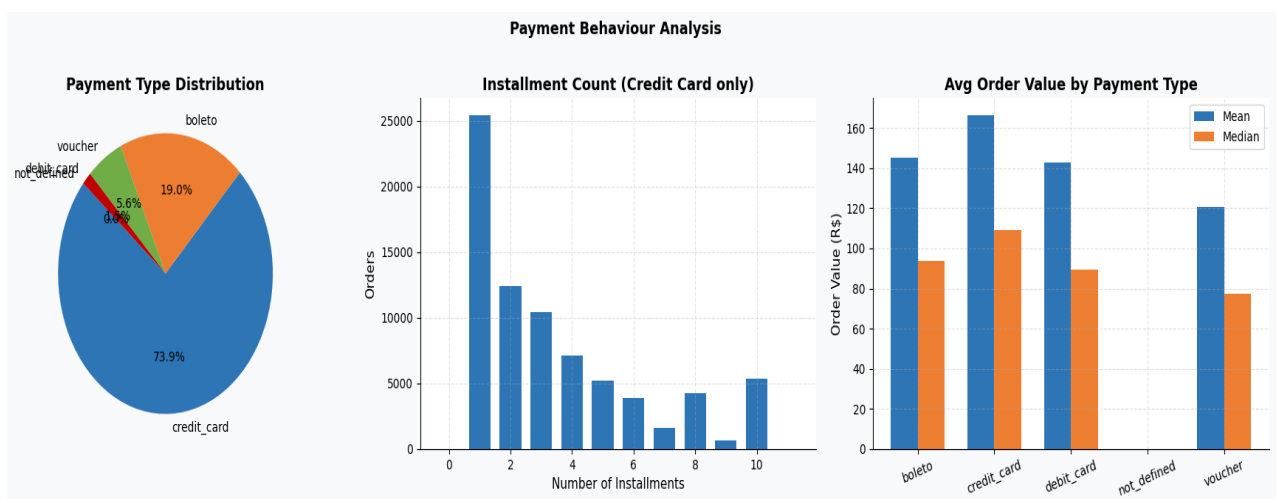
3. Product Category Performance

- Product category analysis shows that segments such as Bed & Bath, Health & Beauty, and Sports & Leisure contribute the majority of total revenue.
- These categories likely benefit from frequent demand and suitability for online purchasing. Interestingly, categories with higher revenue do not always correspond to higher customer satisfaction, indicating potential gaps in product quality or delivery experience.



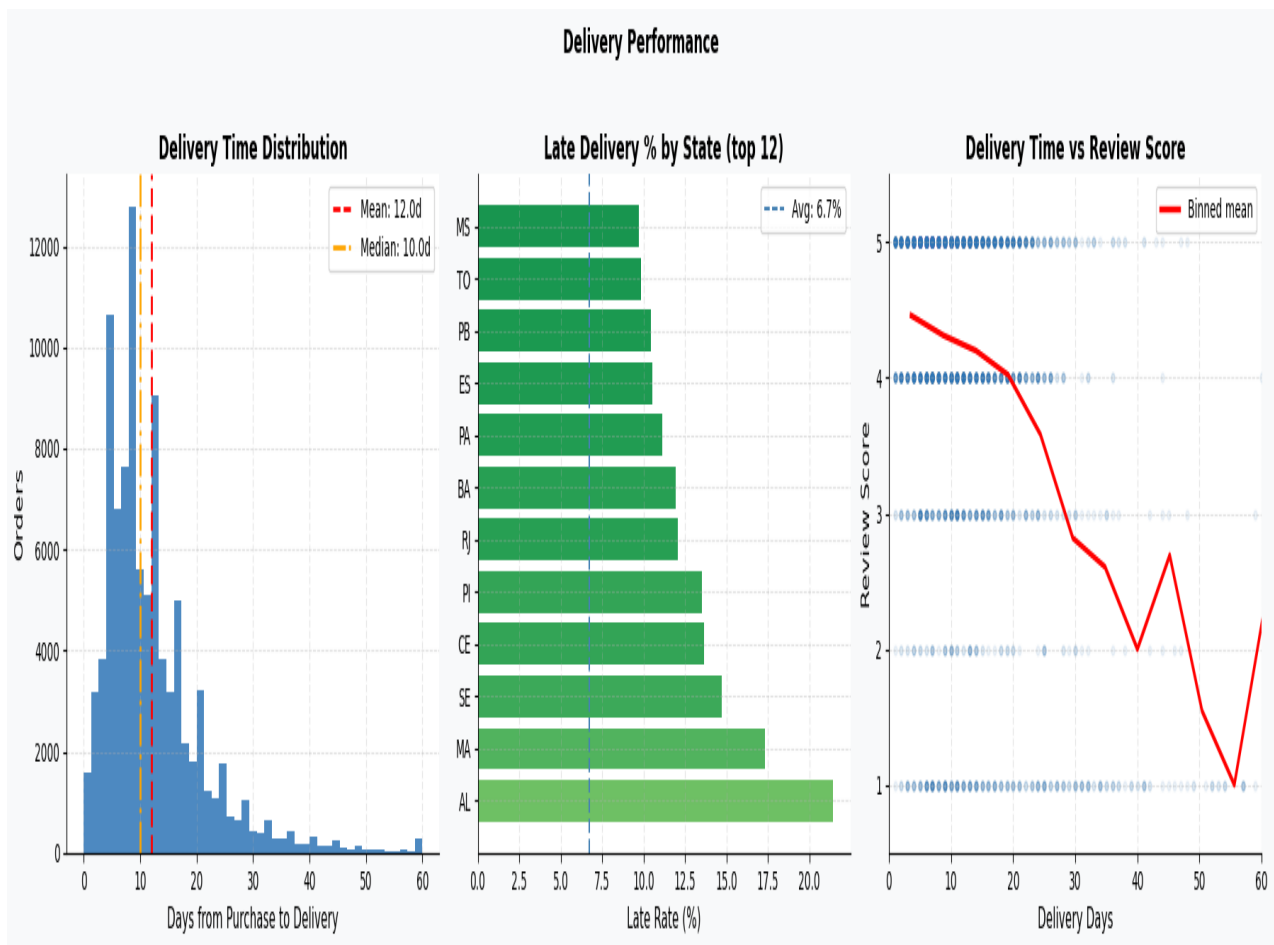
4. Payment Behavior

- **Payment behavior** is **heavily skewed toward credit cards**, which account for approximately 75% of all transactions and are associated with **higher order values compared to Boleto.**
- This reflects strong adoption of installment-based purchasing in the Brazilian market.
- While credit card usage is high, the difference in average order value across payment types is relatively small, suggesting that payment method is more of a convenience factor than a spending constraint.



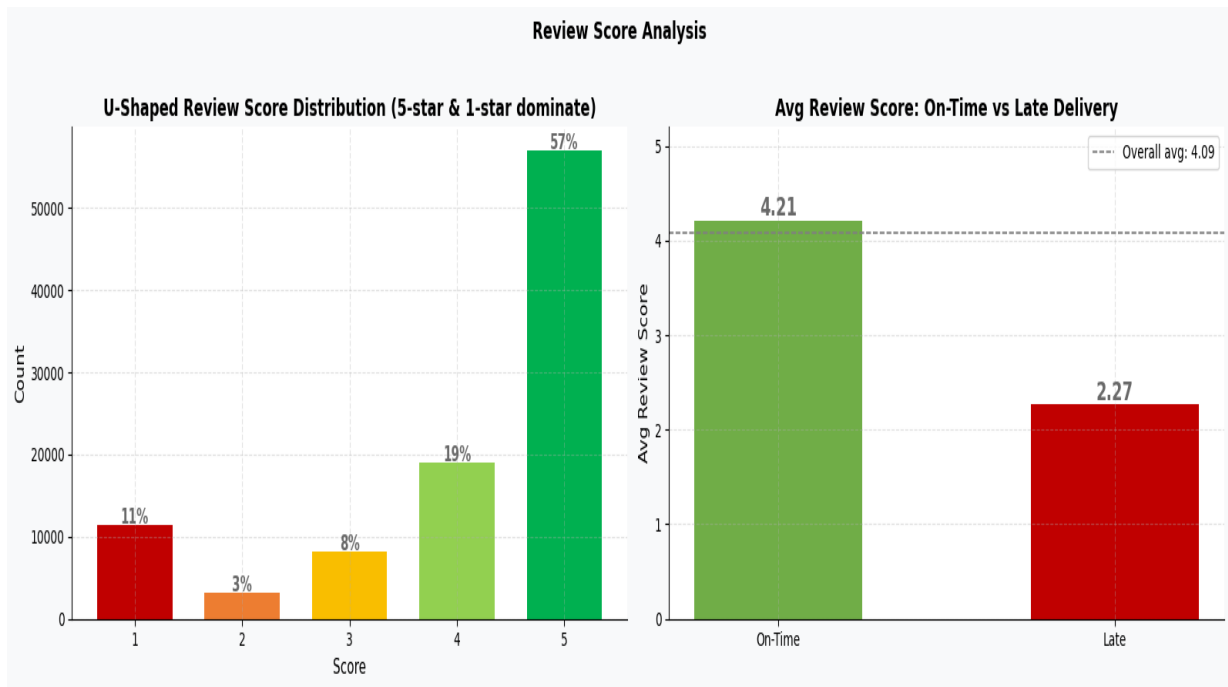
5. Delivery Performance

- Delivery performance remains a critical operational challenge, with approximately 20.7% of orders being delivered later than the estimated date.
- Moreover, delivery times vary significantly across regions, with some states experiencing average delivery durations exceeding 20 days.
- This inconsistency highlights inefficiencies in logistics and last-mile delivery, particularly in high-volume regions.



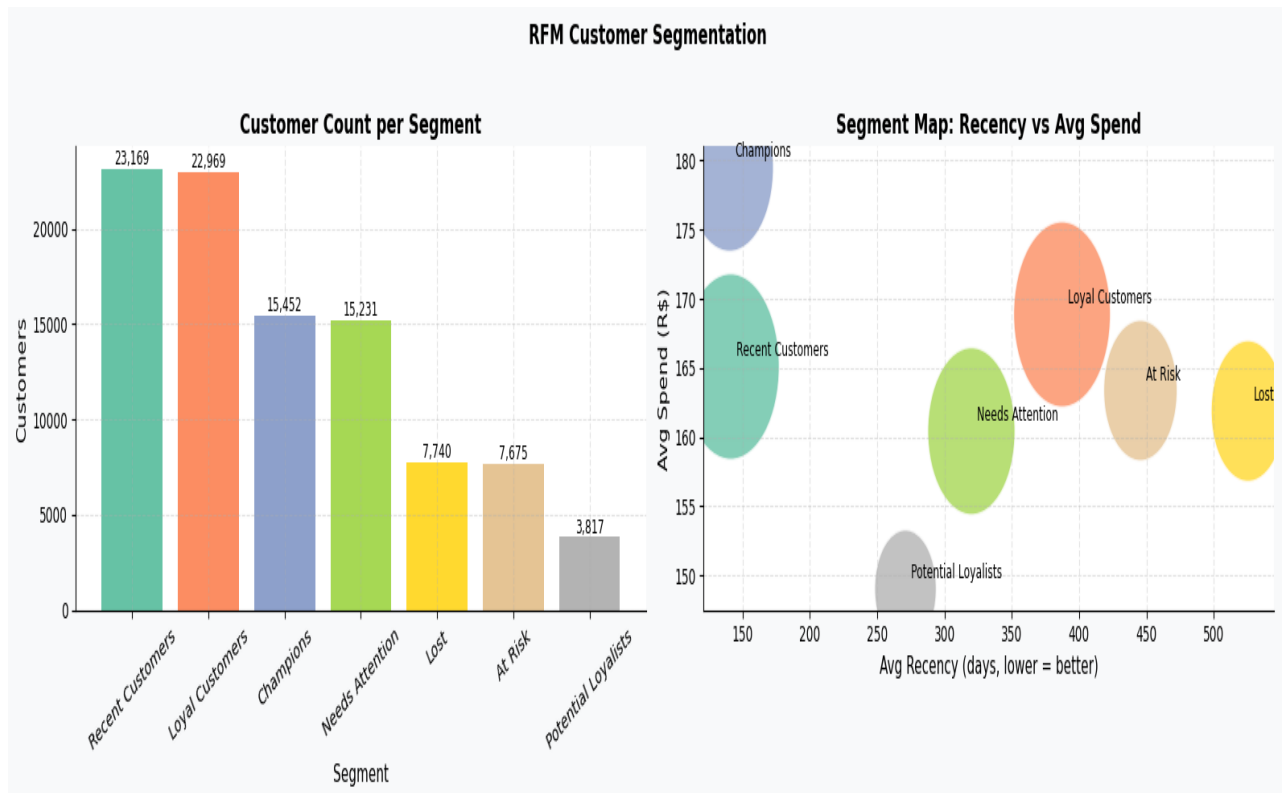
6. Review Score Analysis

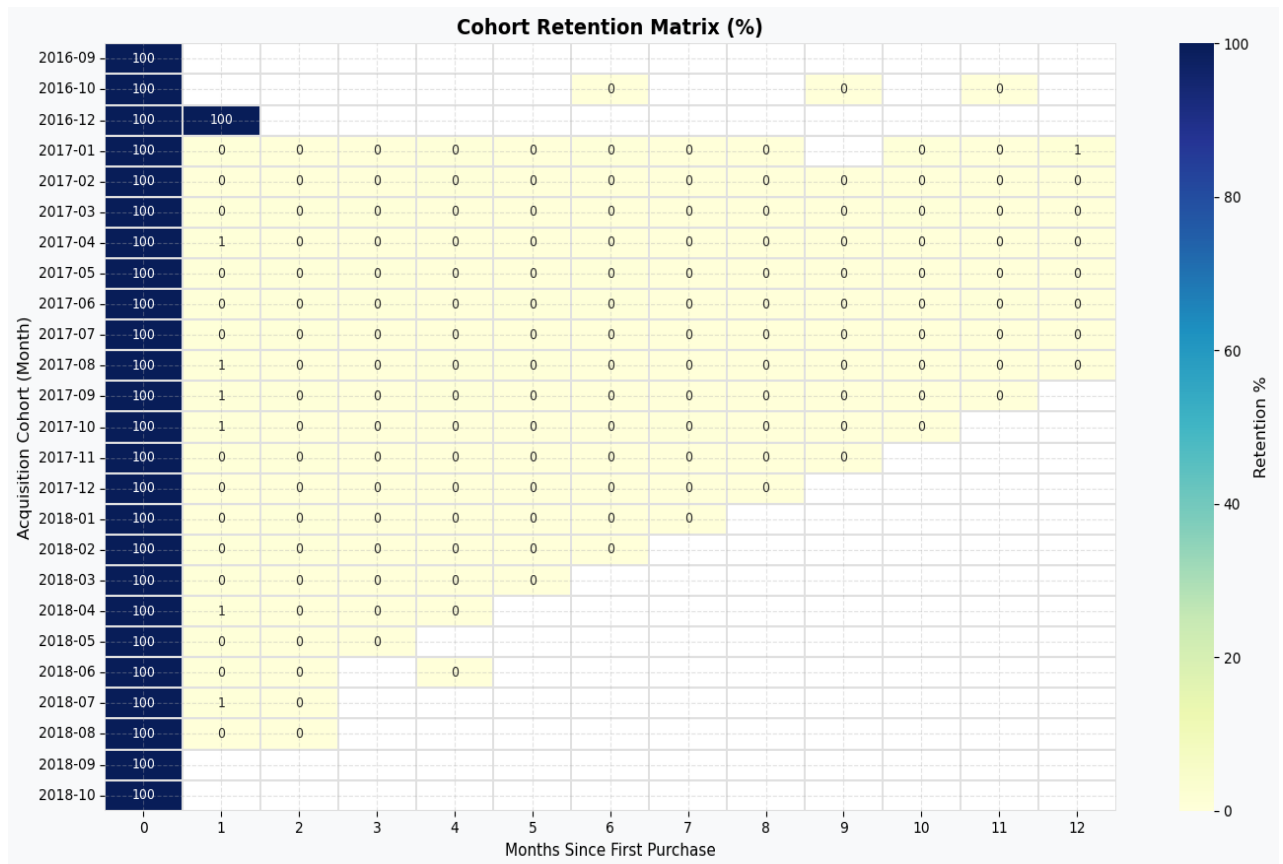
- 5-star reviews dominate, but late deliveries strongly correlate with 1–2star ratings.
- Higher-priced items show slightly different satisfaction patterns (explored further in hypothesis testing).



7. Customer Segmentation (RFM + Cohort)

- Very low repeat purchase rate (3%) indicates weak retention.
- Cohort analysis shows steep drop-off after the first purchase most customers do not return.





HYPOTHESES TESTING

H1: LATE DELIVERIES SIGNIFICANTLY REDUCE REVIEW SCORES

- **LATE ORDERS:** Average review score 3.05
- **ON-TIME ORDERS:** Average review score 4.38
- **MANN-WHITNEY U TEST:** $p\text{-value} < 0.001$ (highly significant)
- **EFFECT SIZE IS LARGE** (1.33point difference).
- **CONCLUSION:** Late delivery is one of the strongest drivers of poor customer experience.

H2: HIGHER-PRICED PRODUCTS RECEIVE DIFFERENT SATISFACTION LEVELS

- Products were divided into price quintiles.
- **Kruskal Wallis test** showed statistically significant differences across price tiers ($p < 0.05$).

- Mid-to-high price items generally performed better in satisfaction than the cheapest segment.
 - Conclusion: **Price** is not just a **revenue factor** it also influences **perceived quality and satisfaction**.
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ACTIONABLE RECOMMENDATIONS

Actionable 1: DELIVERY PROMISE ACCURACY PROGRAM (HIGH PRIORITY)

- Replace static estimated delivery dates with dynamic, ML-based predictions using seller location, customer location, product weight, and historical performance.
- Target $\geq 90\%$ on-time delivery.
- Implement proactive “Sorry + Voucher” compensation for orders likely to be late.
- Create a seller performance scorecard with penalties/rewards for dispatch speed.
- Prioritize logistics fixes in ES, RJ, and DF (high late rates + decent volume).

Actionable 2: BUSINESS-HOURS MARKETING & RETENTION ENGINE

- **Shift all marketing campaigns** (emails, push notifications, flash sales) **to 10am–2pm local time**.
 - Launch **“Lunch Break Deals”** focused on **high-margin, high-satisfaction categories (Health & Beauty, Watches & Gifts)**.
 - **Build a post-purchase retention program:** automated sequences offering complementary products in high-retention categories + loyalty discounts for second purchase within 60 days.
 - Target weekend promotions for higher AOV bundles.
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Key Open Question:

DOES FIRST-ORDER DELIVERY EXPERIENCE PREDICT LONG-TERM CUSTOMER LIFETIME VALUE (LTV)?

WHY IT MATTERS:

If customers who receive their first order on time show 2x higher 90-day retention and repeat purchase rate, then investment in logistics is simultaneously a customer acquisition and retention lever. This would justify significantly higher spend on delivery reliability.

PROPOSED ANALYSIS:

- Cohort-based survival analysis linking first-order `is\_late` flag with repeat purchase behavior.
 - Cross-state delivery route analysis to identify chronic problem corridors.
 - Longer-term data (beyond 2018) would strengthen this analysis.
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Appendix

A. Notebook / Analysis File

The complete analysis notebook, including data cleaning, exploratory data analysis (EDA), feature engineering, hypothesis testing, and visualizations, is available publicly at:

<https://github.com/hasinii12/olist-ecommerce-customer-behavior-analysis>

This ensures full transparency and reproducibility of the analysis. The repository is structured to clearly separate raw analysis, report outputs, and supporting materials.

Due to file size constraints, the original dataset is not included in the repository. It can be accessed from the official Kaggle source:

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Instructions to download and run the notebook are provided within the repository.

B. LLM Usage Disclosure

Models Used:

- ChatGPT (GPT-5.3)
 - Claude (Anthropic – Claude Sonnet)
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Scope of Usage

LLMs were used as assistive tools to improve efficiency and structure during the project. Specifically:

- Supporting Python/pandas syntax refinement
 - Providing templates for statistical tests (Mann-Whitney U, Kruskal-Wallis)
 - Suggesting data visualization improvements (Matplotlib/Seaborn)
 - Assisting in structuring the report and refining explanations
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Not Used For

LLMs were not used for:

- Core data analysis or computation
- Hypothesis formulation
- Interpretation of results
- Business insights and recommendations
- Final decision-making

Author Contribution

All analytical thinking, hypothesis design, interpretation of results, and business recommendations were independently developed. LLMs were used strictly as productivity aids and not as decision-making systems.

Reproducibility Note

The analysis can be fully reproduced by:

1. Downloading the dataset from Kaggle
2. Placing the files in the specified directory (data/)
3. Running the provided JupyterNotebook

This ensures transparency and validation of all findings presented in this report.

End of Report